

L40 — Heaps: Min-Heap and Max-Heap

Unit: 3 | Chapter: Trees | BT Level: BT5 | CO: CO2, CO3

Learning Objectives

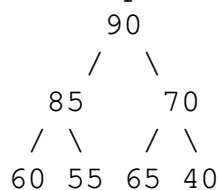
- Define the heap property for min-heaps and max-heaps
 - Implement heap operations: insert, extract, heapify
 - Represent a heap as an array and navigate using index formulas
 - Build a heap from an arbitrary array in $O(n)$ time
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1. Heap Definition

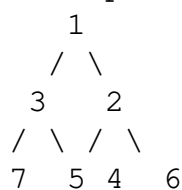
A **heap** is a complete binary tree satisfying the **heap property**:

- **Max-Heap:** Every parent $\geq$ both children $\rightarrow$ root is the maximum.
- **Min-Heap:** Every parent $\leq$ both children $\rightarrow$ root is the minimum.

Max-Heap:



Min-Heap:



Key property: The root is always the max (or min), but siblings have no ordering relationship.

2. Array Representation

Heap nodes are stored level by level in an array (1-indexed):

```
Array: [_, 90, 85, 70, 60, 55, 65, 40]
Index:   1   2   3   4   5   6   7
```

For node at index i :

- **Parent:** $i // 2$
- **Left child:** $2 * i$
- **Right child:** $2 * i + 1$

(0-indexed: parent = $(i-1) // 2$, left = $2i+1$, right = $2i+2$)

3. Max-Heap Implementation

```
class MaxHeap:
```

```

def __init__(self):
    self.heap = []    # 0-indexed

def _parent(self, i): return (i - 1) // 2
def _left(self, i): return 2 * i + 1
def _right(self, i): return 2 * i + 2

def _swap(self, i, j):
    self.heap[i], self.heap[j] = self.heap[j], self.heap[i]

def peek(self):
    """Return max without removing. O(1)"""
    if not self.heap:
        raise IndexError("Heap is empty")
    return self.heap[0]

def insert(self, val):
    """Insert value and restore heap property. O(log n)"""
    self.heap.append(val)
    self._sift_up(len(self.heap) - 1)

def _sift_up(self, i):
    """Bubble element up until heap property restored."""
    while i > 0:
        parent = self._parent(i)
        if self.heap[i] > self.heap[parent]:
            self._swap(i, parent)
            i = parent
        else:
            break

def extract_max(self):
    """Remove and return max element. O(log n)"""
    if not self.heap:
        raise IndexError("Heap is empty")
    if len(self.heap) == 1:
        return self.heap.pop()

    # Swap root with last, remove last, sift down root
    max_val = self.heap[0]
    self.heap[0] = self.heap.pop()    # Move last element to root
    self._sift_down(0)
    return max_val

def _sift_down(self, i):
    """Push element down until heap property restored."""
    n = len(self.heap)
    while True:
        largest = i
        left = self._left(i)
        right = self._right(i)

        if left < n and self.heap[left] > self.heap[largest]:

```

```

        largest = left
    if right < n and self.heap[right] > self.heap[largest]:
        largest = right

    if largest != i:
        self._swap(i, largest)
        i = largest
    else:
        break

```

4. Insertion Trace

Insert 80 into Max-Heap: [90, 85, 70, 60, 55, 65, 40]

1. Append 80 at index 7: [90, 85, 70, 60, 55, 65, 40, 80]
2. Sift up: index 7, parent = 3 (value 60); $80 > 60 \rightarrow$ swap: [90, 85, 70, 80, 55, 65, 40, 60]
3. Sift up: index 3, parent = 1 (value 85); $80 < 85 \rightarrow$ stop.

Final: [90, 85, 70, 80, 55, 65, 40, 60] ✓

5. Extract-Max Trace

Extract max from [90, 85, 70, 80, 55, 65, 40, 60]:

1. Save max = 90
2. Move last (60) to root: [60, 85, 70, 80, 55, 65, 40]
3. Sift down from index 0:
 - Compare 60 with children: left=85 (idx 1), right=70 (idx 2). Largest=85 $\rightarrow$ swap: [85, 60, 70, 80, 55, 65, 40]
 - At index 1: left=80 (idx 3), right=55 (idx 4). Largest=80 $\rightarrow$ swap: [85, 80, 70, 60, 55, 65, 40]
 - At index 3: children are at idx 7,8 — don't exist. Stop.

Final: [85, 80, 70, 60, 55, 65, 40] ✓ (85 is new max)

6. Build Heap from Array — $O(n)$

Naive approach: Insert n elements one by one $\rightarrow O(n \log n)$.

Optimal approach (Heapify): Start from last non-leaf, sift down each node.

```

def build_max_heap(arr):
    """Build max-heap in-place in  $O(n)$  time."""
    n = len(arr)

    def sift_down(heap, i, size):
        while True:

```

```

        largest = i
        left, right = 2*i+1, 2*i+2
        if left < size and heap[left] > heap[largest]:
            largest = left
        if right < size and heap[right] > heap[largest]:
            largest = right
        if largest != i:
            heap[i], heap[largest] = heap[largest], heap[i]
            i = largest
        else:
            break

    # Start from last internal node
    for i in range(n // 2 - 1, -1, -1):
        sift_down(arr, i, n)

    return arr

arr = [4, 10, 3, 5, 1, 9]
print(build_max_heap(arr))    # [10, 5, 9, 4, 1, 3]

```

Why $O(n)$?

Most nodes are near the bottom and require little sifting. The cost is: $\sum_{h=0}^{\lfloor \log n \rfloor} \frac{n}{2^{h+1}} \cdot h = O(n)$

7. Min-Heap (via heapq)

Python's `heapq` module implements a **min-heap**:

```

import heapq

heap = []
heapq.heappush(heap, 5)
heapq.heappush(heap, 3)
heapq.heappush(heap, 7)
heapq.heappush(heap, 1)

print(heap)                # [1, 3, 7, 5] (min-heap)
print(heapq.heappop(heap))  # 1 (min)
print(heapq.heappop(heap))  # 3

# Build heap from list
arr = [4, 10, 3, 5, 1, 9]
heapq.heapify(arr)          #  $O(n)$  – converts in place
print(arr)                  # [1, 4, 3, 5, 10, 9]

```

8. Complexity Summary

Operation	Time	Notes
peek (min/max)	$O(1)$	Root access

insert	$O(\log n)$	Sift up at most h levels
extract-min/max	$O(\log n)$	Sift down at most h levels
build heap	$O(n)$	Linear time heapify
search	$O(n)$	Heap has no ordering except root
delete (arbitrary)	$O(\log n)$	Replace with last, sift

Summary

- **Heap:** complete binary tree with parent $\geq$ children (max-heap) or parent $\leq$ children (min-heap).
 - Array representation: parent at i , children at $2i + 1$ and $2i + 2$ (0-indexed).
 - **Insert:** append, sift up — $O(\log n)$.
 - **Extract:** swap root with last, remove, sift down — $O(\log n)$.
 - **Build heap:** start from last internal node, sift down — $O(n)$.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 6 (MIT Press, 2022)
- Python `heapq` module documentation